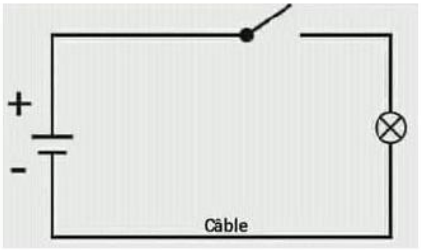
Electronics Laboratory

First meeting

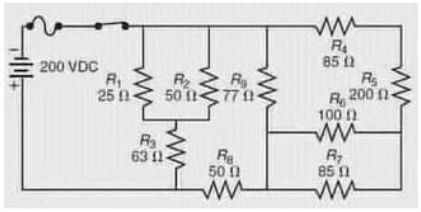
„Electronics are simple!”

Electronics

at the lecture



at the exam



and in real life



Staff

Head of laboratory: Viktor Farkas (viktor.farkas@reak.bme.hu)

Students:

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- Imre Harmos (imre.harmos@gmail.com)
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- András Zsákai (zsakai.andras@ek.hun-ren.hu)

Our measurements

- Transistor amplifiers and the Miller-effect (M1)
- Basic circuits for operational amplifiers (M2)
- Comparators (M3)
- Digital circuits (M4)
- Flip-flops and soldering (M5)
- Microcontroller programming in C (M6)

The measurement guides are available on this website:
https://oktatok.reak.bme.hu/farkas/electronics-laboratory/

Laboratory requirements

- Laboratory courses will take place in classroom 214. in building R (Institute of Nuclear Techniques)
- Knowledge of the measurement instructions is mandatory.
- Exam will be a multiple choice test.

  (with a drawing assignment)
- Report (for next measurement).
- Late submission of report will result in deduction of a mark!
- All measurements must be taken, 1 can be made up.

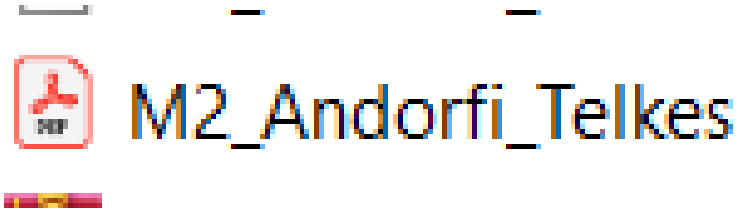
Laboratory reports

- **Name, Neptun code, e-mail address, name of measurement, date must be indicated.**
- The reports have to be uploded to Moodle group.
- Name of the report:

  MX_LastName1_LastName2.pdf

  (X is the number of measurement)
- Every report worth **10 points**. The point deductions and plus points are justified by the corrector and sent to the measurement pair.

Points	Grade
0-2	1
3-4	2
5-6	3
7-8	4
9-10	5





Budapest University of Technology and Economics
Faculty of Natural Sciences

M2 - Basic circuits for operational amplifiers

Lab report

04/09/2026

Lab responsible
Viktor Farkas

Student
István Andorfi (THEDAW)
PhD in Physics 3rd grade
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Student
Ágota Telkes (FLD9PP)
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Electronics laboratory

Point deductions / plus points 1

- There is a 0.5 points deduction if
  - Format of the report is not a pdf
  - Name of the file is not in the given form
  - Many spelling mistakes
  - Aesthetically unappealing report
  - Too long (12< pages) or too short (<3 pages)
  - List of applied instruments is incomplete or incorrect
  - Sign errors

Point deductions / plus points 2

- There is a 1 point deduction if
  - List of applied instruments is missing
  - Incorrect units or orders of magnitude
  - Error greater than $\pm 30\%$ without a justification
  - Unfinished measurement (by lack of time), which is communicated in the report
  - No comparison of measurement results with theory
  - No captions of figures and tables
  - Axes of diagrams without labels and units
  - Missing measurement data at multiple step exercises
  - No photos (if they are required)
  - Adjectives instead of numbers (big, small, fast, etc)

Point deductions / plus points 3

- There is a 2 points deduction if
  - Missing exercise (without any measurement results)
  - Handdrawn diagrams
  - Missing complete description of the exercise
- There is 1 plus point if
  - Measuring the exact values of the circuit elements (resistances, etc) and use these in the evaluation
  - Substantive, aesthetically pleasing, high-quality report

About lab reports

Philosophy of lab reports

- Laboratory reports model the world of scientific articles, monographs, and reports about real measurements.
- Consequently similar formal requirements are true for a lab report as for any scientific writing.
- Lab reports help you to get familiar with scientific writing and to prepare for future publications and theses corresponding to your studies.
- The aim of a measurement report ensuring repeatability, say if someone from Princeton reads your report shall be able to perform the measurement and get the same results.

Structure of a scientific writing

- Abstract: what is the main focus of the article.
- Introduction: how the presented work fit in the previous results.
- Theoretical background: what previous results heve we used.
- Methods: how we got our results.
- Results: what we got, and how it is important.
- Summary, conclusion, outlook: summarizing what we got, and what can be the next step.
- Acknowledment
- Bibliography

Structure of scientific reading

- Title: hm, that seems interesting!
- **Abstract**: am I interested in this topic?
- **Summary**: what did they got?
- Results: did they really got, what they told?
- Methods: how the hell did they get that?
- Theoretical background: where these equations come from?
- Introduction: have they read anything that I know?

In general about scientific work

- It happens sometimes, that your measurement is wrong. Write down, that it is worng!
- Always think about your work as what would you feel if you were the reader!
- Try to be precise and demanding!
- Your aim shall be to teach what you've done to the reader. Be informative!
- Before you start to solve an exercise, please read and interpret it!
- If you face some unexpected behaviour tell the reader, that here is something unexpected, and try to explain!

How to measure?

- We have theoretical expectations (never measure without expectations).
- Perform a measurement to confirm these expectations.
- I've observed something:
  - Is it compatible with my expectations (between ranges of error)?
  - If yes, I can write it down.
  - If not, think about why! Did I underestimate the error? Did I do something wrong? Is my theory incorrect?

Some general formal requirements:

- Avoid half-page long empty spaces! Don't leave empty sections!
- Non-index type subscripts and units shall be roman, units shall be separated from the number: $R_s = 100k\Omega$ or $R_{\mathrm{s}} = 100\ \mathrm{k\Omega}$.
- Before sending in, try to look for typical errors (?? in a LaTeX document). Always peruse the final version!
- If you have a few data $(\sim < 20)$ write it to a table!
- If you write down something try to confirm it with some data, plot or picture.
- If you refer to something, please refer to it correctly!
- Try to use „scientific language".

Requirements for figures

- All figures shall have a number and a caption. (e.g. Figure 2.: Something intersting).
- Export the figures correctly!
- All figures and tables shall be comprehensible by its own.
- Always name the axes, tell what is on the figure!
- Always highlight the measured data points!

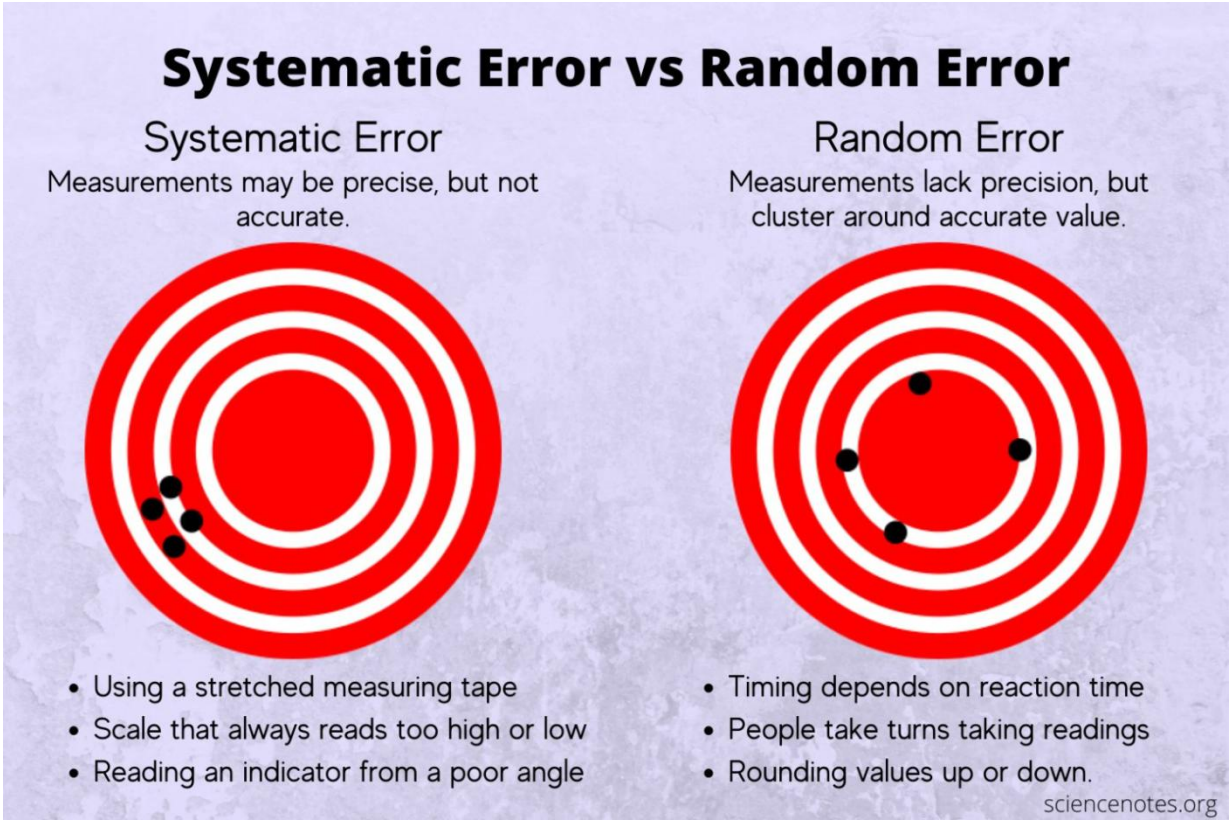
Basics of error estimation

Basics of error estimation

- Error is the difference between the measured value and **reality**.
- Types of error:
  - Systematic error:
    - Instrumental
    - Enviromental
    - Observational
    - Theoretical
  - Random error
- Handling:
  - Systematic: change setup
  - Random: repeat

Basics of error estimation





Example: summing amplifier

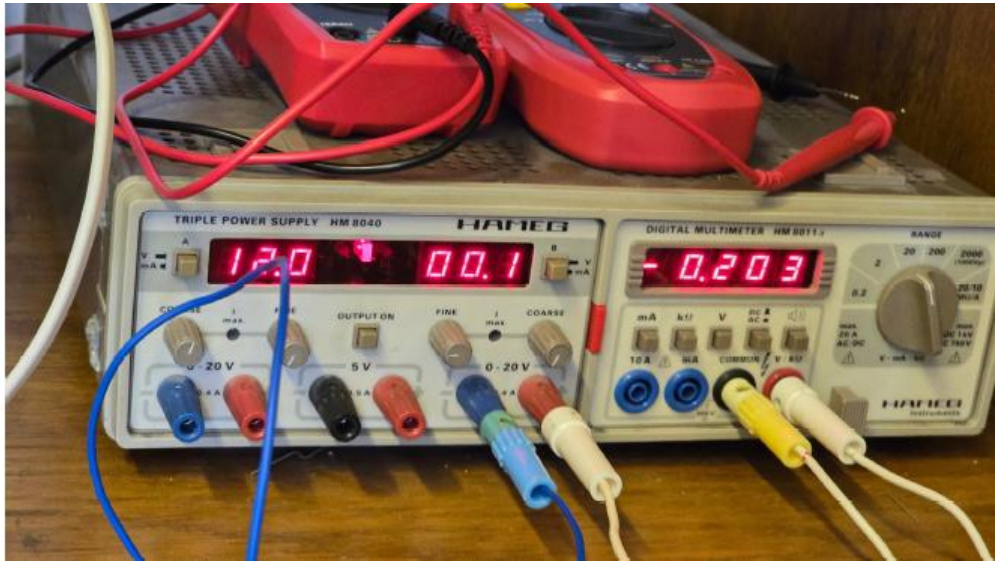
- Using HAMEG HM 8040 as a power supply and HAMEG HM 8011 as a multimeter.

$$U_{\text{out}} = -\left(\frac{R_{\text{v}}}{R_1}U_1 + \frac{R_{\text{v}}}{R_2}U_2\right)$$

Error sources:

- Resistance values
- Voltage measurement

- Theoretical assumptions (zero resistance for cables, no current to the amplifier)
- Non-ideal behaviours



The above equation shall hold between ranges of error.

Solution: Gaussian error propagation:

If for physical quantity W

$$W = f(x_1, x_2, \ldots, x_n)$$

Then

$$\Delta W = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \Delta x_i \right)^2}$$

- Let the error of resistances be $\frac{\Delta R}{R} = \pm 5\%$
- Let us consider $U_1 = U_2 = U = 0.1\ \mathrm{V}$
- Then the error of the right hand side:
- $\Delta U_{\mathrm{out}}^{\mathrm{rhs}} = \sqrt{U_{\mathrm{out}}^2 \left(\frac{\Delta R}{R}\right)^2 + 2U^2 \left(\frac{\Delta R}{R}\right)^2 + 2U^2 \left(\frac{\Delta U}{U}\right)^2}$
- Since $U_{\mathrm{out}} \approx -2U$:
- $\frac{\Delta U_{\mathrm{out}}^{\mathrm{rhs}}}{U_{\mathrm{out}}} \approx \sqrt{\frac{3}{2}\left(\frac{\Delta R}{R}\right)^2 + \frac{1}{2}\left(\frac{\Delta U}{U}\right)^2}$
- How much is $\frac{\Delta U}{U}$?
- You can set set U by a $0.1\ \mathrm{V}$ precision $\Rightarrow\ \Delta U = \pm 0.05\ \mathrm{V} \Rightarrow \left|\frac{\Delta U}{U}\right| = 50\%$
- Then $\frac{\Delta U_{\mathrm{out}}^{\mathrm{rhs}}}{U_{\mathrm{out}}} \approx \sqrt{\frac{3}{2}0.05^2 + \frac{1}{2}0.5^2} = 36\%$, so $U_{out} \in [-0.272\ \mathrm{V}; -0.128\ \mathrm{V}]$ is acceptable.

- But on the left hand side:
- You can measure U_{out} in $0.001\ \mathrm{V}$ precisely, say $\Delta U_{\mathrm{out}}^{\mathrm{lhs}} = \pm 0.0005\ \mathrm{V}$, so
- $\left|\frac{\Delta U_{\mathrm{out}}^{\mathrm{lhs}}}{U_{\mathrm{out}}}\right| = 0.5\%$

- Consequently: error is dominated by the uncertainty of the input voltage, the theory shall be correct in this error ranges!
- This is a systematic error, can be handled by the direct measurement of the voltage coming from the power supply (reduce $\Delta U/U$ to 0.5%).
- Then $\frac{\Delta U_{\mathrm{out}}^{\mathrm{rhs}}}{U_{\mathrm{out}}} \approx \sqrt{\frac{3}{2}0.05^2 + \frac{1}{2}0.005^2} = 6\%$, so $U_{out} \in [-0.212\ \mathrm{V}; -0.188\ \mathrm{V}]$ is acceptable.

Parameter estimation – curve fitting

- Exact mathematics are complicated (Mathematical Methods of Experimental Data Evaluation course with Dr. Réka Szilvási).
- Believe: curve fitting is the most accurate way to estimate the value of a parameter! Do not average measurement data!
- n parameters can be exactly fitted on n measurement points, but it is basically not the way we work (von Neumann's elephant).
- Curve fitting is automatized by computers (Matlab, Python, Wolfram, IGOR, etc.), but be careful!

Our devices

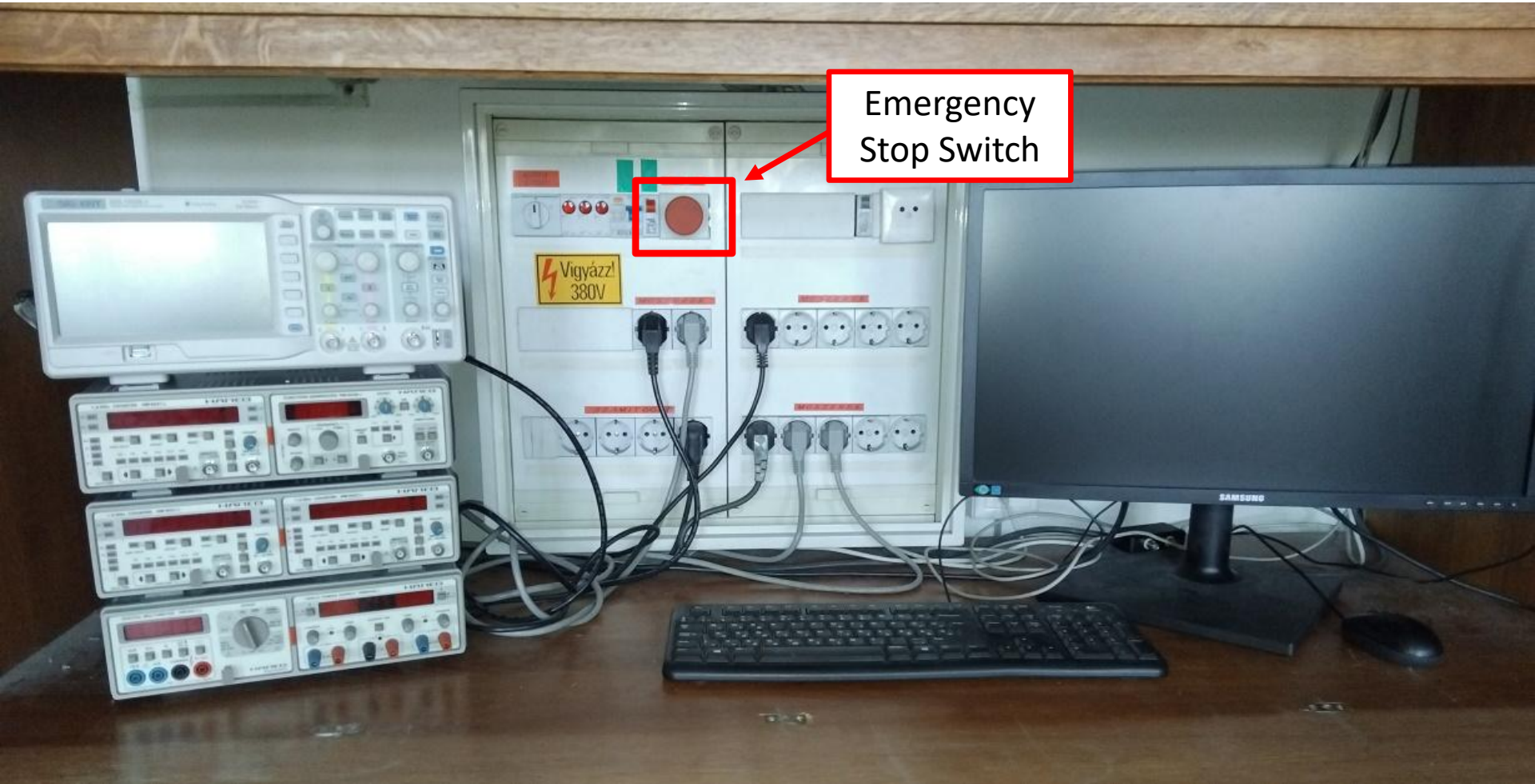
Every measuring station has:

- 1 handheld multimeter
- 1 bench multimeter
- 1 digital oscilloscope
- 1 function generator
- 1 DC power supply with 3 ground-independent outputs
- 1 parts box
- 1 breadboard
- test leads, crocodile clips, BNC T-splitters



Accident prevention

Disconnecting workplaces from the power supply using an emergency stop switch.



Accident prevention

Other hazards in our work

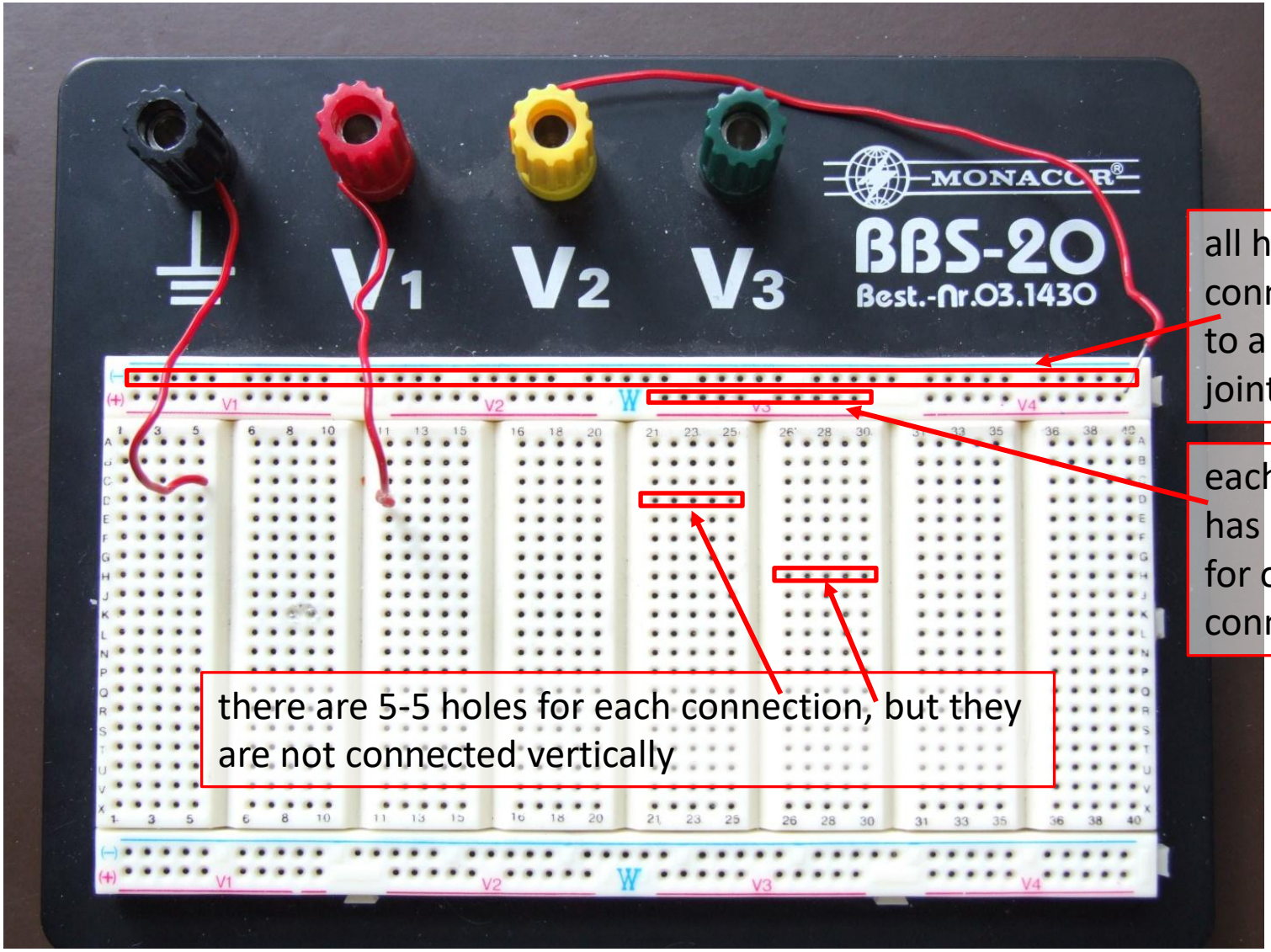
- The measuring tip of the oscilloscope probe is very sharp due to good contact.
- When soldering, the temperature of the soldering iron is approx. 300 °C.
- The solder is a 63/37% tin-lead alloy. Lead is a heavy metal and is toxic! → Wash your hands after work and before eating!







Plug-in test panel (breadboard)

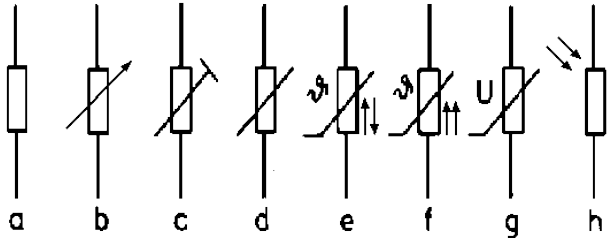


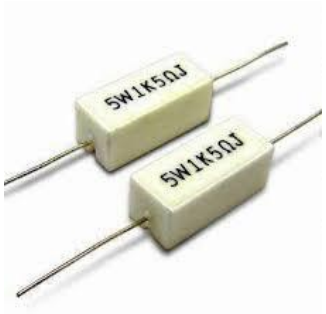
Resistors value marking

With captions or color codes.



Value marking: R=Ohm (Ω), k=kilo Ω, M= mega Ω
Examples of captions:
4R7=4,7 Ω, 68k=68 kΩ, 2M2=2,2 M Ω, k68=680 Ω

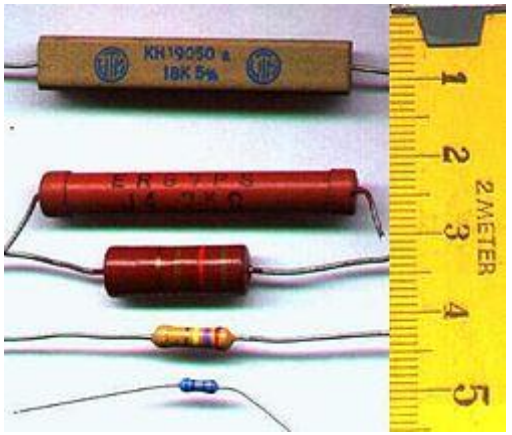




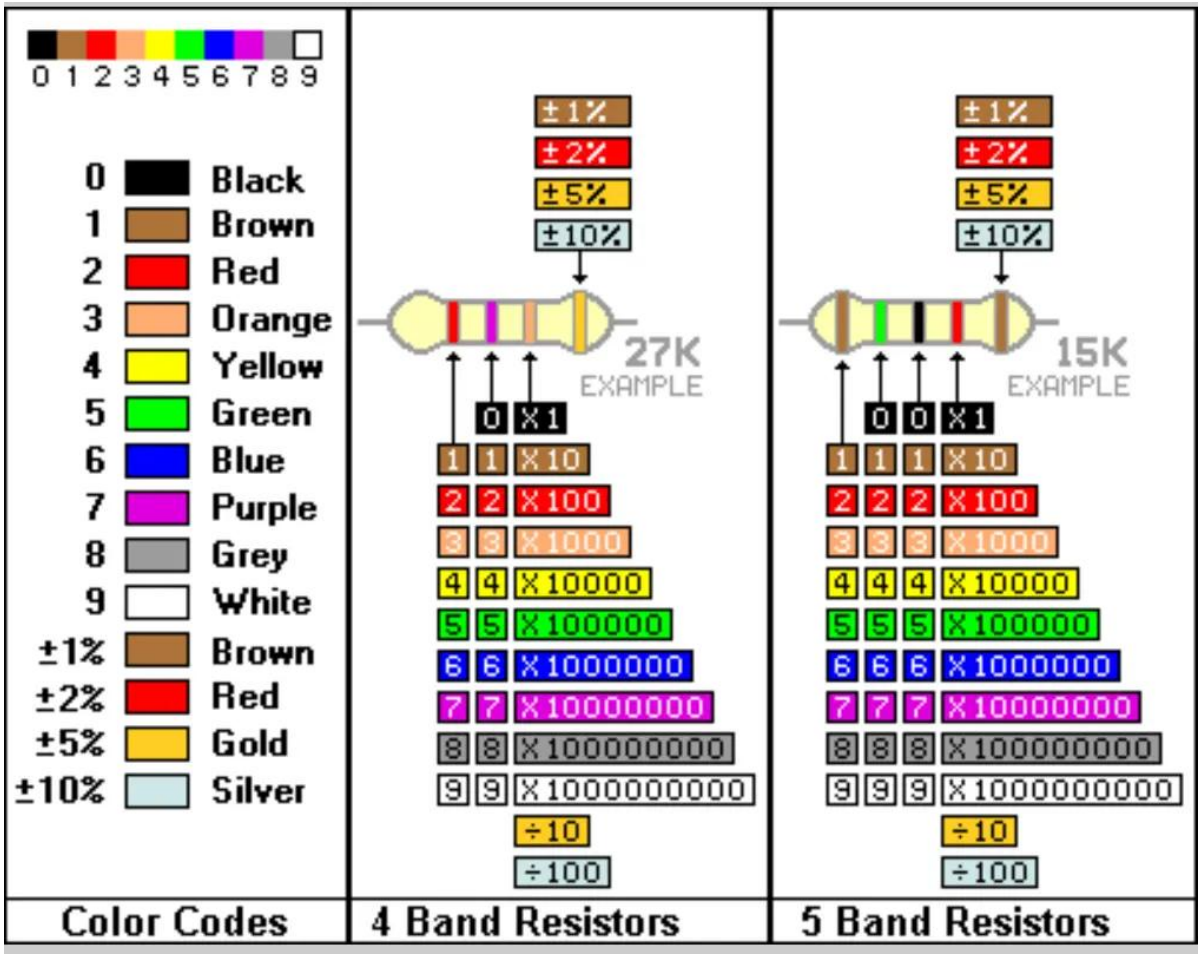


Color codes can be 4 or 5 bands.





Resistors value marking with color codes



Capacitors value marking



For most electrolytic capacitors, the capacitance value is given in uF or mF.
For film capacitors, this value is given in uF or nF.



For some tantalum electrolytic capacitors and ceramic capacitors, the capacitance value is indicated as follows.

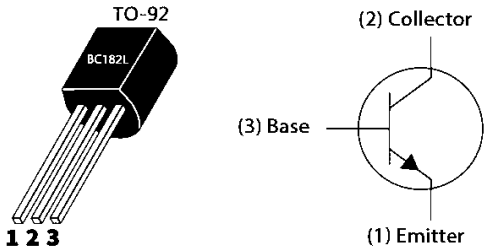
The value is given in pF (10^{-12}), followed by an exponent.

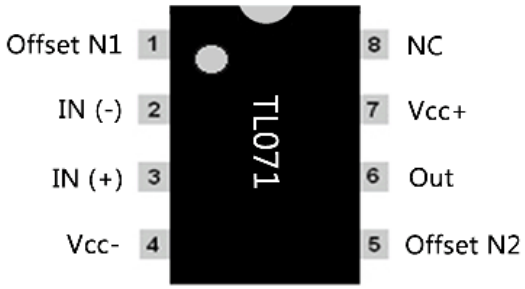
Example:
$103 = 10 \cdot 10^3 = 10\,000$ pF = 10 nF

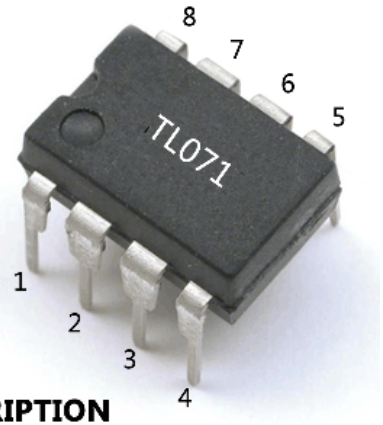
$474 = 47 \cdot 10^4 = 470\,000$ pF = 470 nF

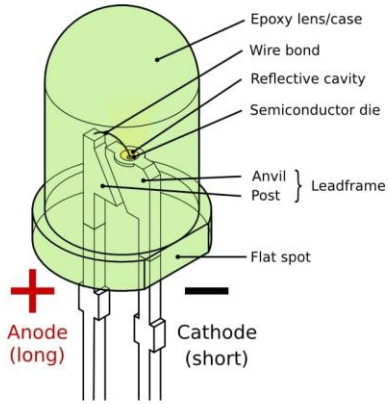
Connecting semiconductors

BC182 Pinout

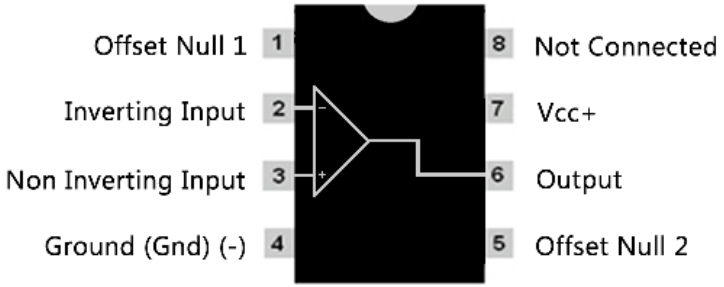








TL071 DETAILED PIN DESCRIPTION



Troubleshooting (circuit activation)

- Voltage measurement in the correct order:
  - Supply voltage(s) value
  - Input voltage(s) value
  - Output voltage(s) value
  - Do the voltages match the expected (!) values?
- Checking contacts
- If all of the above are OK, a component fault may be suspected. (In this case, it is advisable to contact the measurement supervisor for assistance.)

Measurement dates

Measurements are held for 4x45 minutes combined, but you can go out for a break at any time.

Accordingly, morning classes will be held from
T1: 8:15 to 11:15,

afternoon classes from
T2, T4: 12:15 to 15:15,

and evening classes from
T3, T5: 16:15 to 19:15,

The schedule for the measurements, including the times of the classes and the measurement descriptions, will be posted on Moodle.

Questions

Thank you for your attention!
Do you have any questions?